

SPRING BOARD VIRTUAL INTERSHIP 6.0

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Domain : Data Visualization

PROJECT : VISUALIZING U.S NATURAL DISASTER DECLARATIONS

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Milestone – 3 : Geographical Distribution Analysis

Objective:

The goal of this milestone is to **analyze and visualize the geographical distribution of natural disaster declarations across the United States**.

Interns will explore how disaster declarations vary by **state and region**, and identify **geographical hotspots** where disasters occur more frequently.

This stage focuses on converting aggregated data into **maps and spatial visualizations** to reveal patterns that cannot be easily observed in tabular data.

According to the project plan, this milestone involves **aggregating data geographically and creating maps such as choropleth visualizations and state-level comparisons** visualizing US Natural disaster.

Dataset Used:

The dataset used for this project is obtained from the **Federal Emergency Management Agency (FEMA)** disaster database. FEMA provides publicly available data related to disaster declarations and emergency management within the United States.

This dataset contains historical records of disasters and emergencies, including information about their location, type, and declaration details.

Key Attributes in the Dataset

The dataset includes the following important fields:

- **State** – U.S. state where the disaster occurred
- **Declaration Date** – Date when the disaster was officially declared
- **Disaster Type** – Type of disaster (Flood, Hurricane, Fire, Storm, etc.)
- **Incident Type** – Category of the disaster event

We have taken the cleaned dataset from milestone 1.

This dataset helps in analyzing **patterns of disasters across different states and time periods**, enabling better understanding of disaster frequency and regional vulnerability.

Environment Setup:

The project was developed using **Visual Studio Code** with **Jupyter Notebook** support and the **Python** programming language. This environment enables interactive data analysis, visualization, and easy execution of Python code.

Several Python libraries were used for data processing, visualization, and geospatial analysis. **Pandas** and **NumPy** were used for data manipulation and aggregation. For visualization, **Matplotlib**, **Seaborn**, and **Plotly** were used to create charts and interactive plots. For geospatial mapping and spatial analysis, **GeoPandas** and **Folium** were used to generate maps such as choropleth visualizations.

Data Exploration:

Before performing geographical analysis, the dataset was explored to understand its structure, variables, and data distribution. The cleaned dataset `usnd_cleaned.csv` was loaded using Pandas in Python.

Initial exploration involved examining the dataset structure, checking column names, and reviewing the first few records. Key columns such as `state`, `incidentType`, `year`, and `month` were analyzed to understand disaster declaration patterns.

Basic exploratory data analysis (EDA) steps included:

- Inspecting dataset shape and column information
- Checking for missing or inconsistent values
- Identifying the unique states present in the dataset
- Exploring different disaster types recorded in the data
- Reviewing the distribution of disaster declarations over time

This exploration helped ensure that the dataset was suitable for geographical aggregation and visualization, enabling further analysis such as state-level disaster counts and incident-type comparisons.

Core Visualization:

1.Chloropeth map:

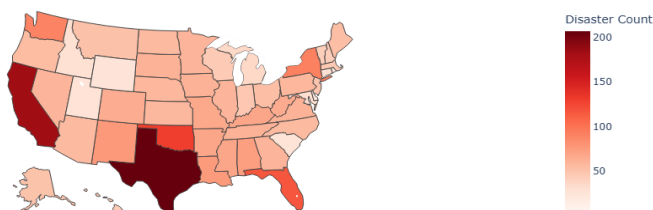
Chart Type

Geographical Choropleth Map

Description

A choropleth map displays disaster counts using color intensity across states. Darker colors represent states with higher disaster frequency.

Geographical Distribution of Natural Disaster Declarations Across the United States



What the visualization represents

The choropleth map represents the distribution of natural disaster declarations across different states in the United States. Each state is colored according to the number of disaster declarations it has experienced, with darker shades indicating higher disaster counts and lighter shades indicating lower counts.

What spatial pattern is observed

The map shows that Texas has the highest number of disaster declarations, followed by states such as California and Florida. Southern and coastal states generally display darker colors, indicating a higher concentration of disasters compared to northern and inland states.

pattern

The higher disaster frequency in states like Texas, Florida, and California can be explained by their geographical location and climatic conditions. Coastal states are more vulnerable to hurricanes, floods, and tropical storms, while western states such as California frequently experience wildfires and drought-related disasters. These environmental factors contribute to the higher number of disaster declarations in these regions.

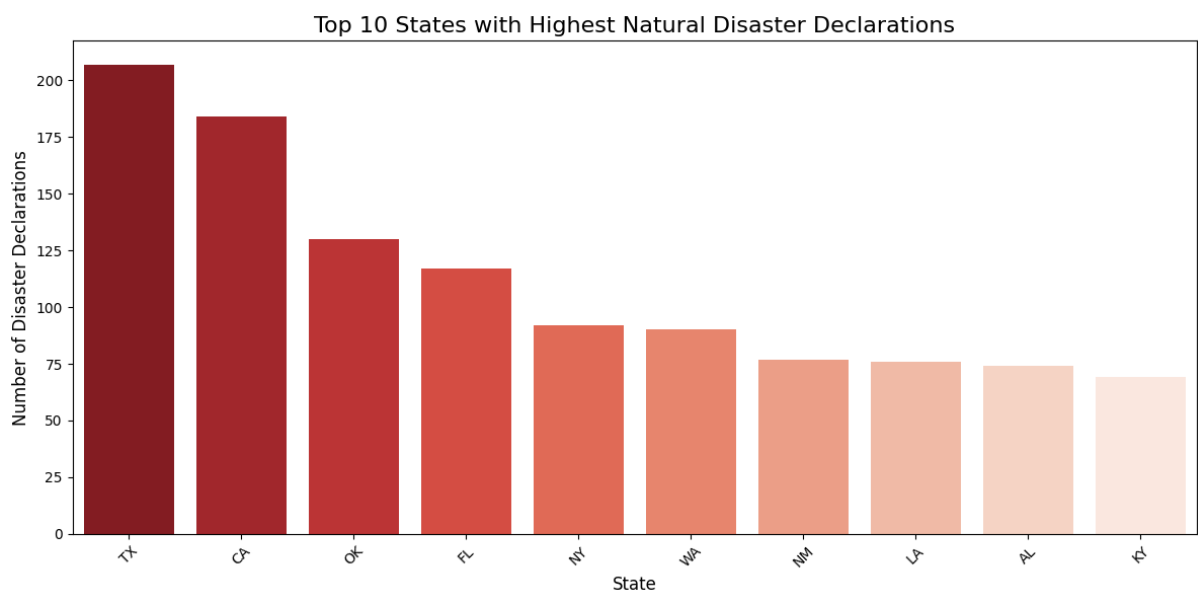
State Comparison Chart:

Chart Type

Bar Chart

Description

Compare disaster counts across the top states. Focus on the **Top 10 states with highest declarations.**



What the visualization represents

The bar chart represents the top 10 states in the United States with the highest number of natural disaster declarations. Each bar corresponds to a state, and the height of the bar indicates the total number of disaster declarations recorded in that state.

What spatial pattern is observed

The chart shows that Texas has the highest number of disaster declarations, followed by California and Oklahoma. Several southern and central states such as Florida, Louisiana, Alabama, and Kentucky also appear among the top disaster-prone states. This indicates that disasters are more frequent in certain regions rather than being evenly distributed across all states.

Possible explanation for the pattern

The higher number of disasters in states like Texas, California, Florida, and Oklahoma can be attributed to their geographical and climatic conditions. Southern states frequently experience hurricanes, floods, and severe storms, while western states such as California are more vulnerable to wildfires and drought-related disasters. Additionally, central states like Oklahoma are often affected by severe storms and tornadoes, which contributes to their high disaster declaration counts.

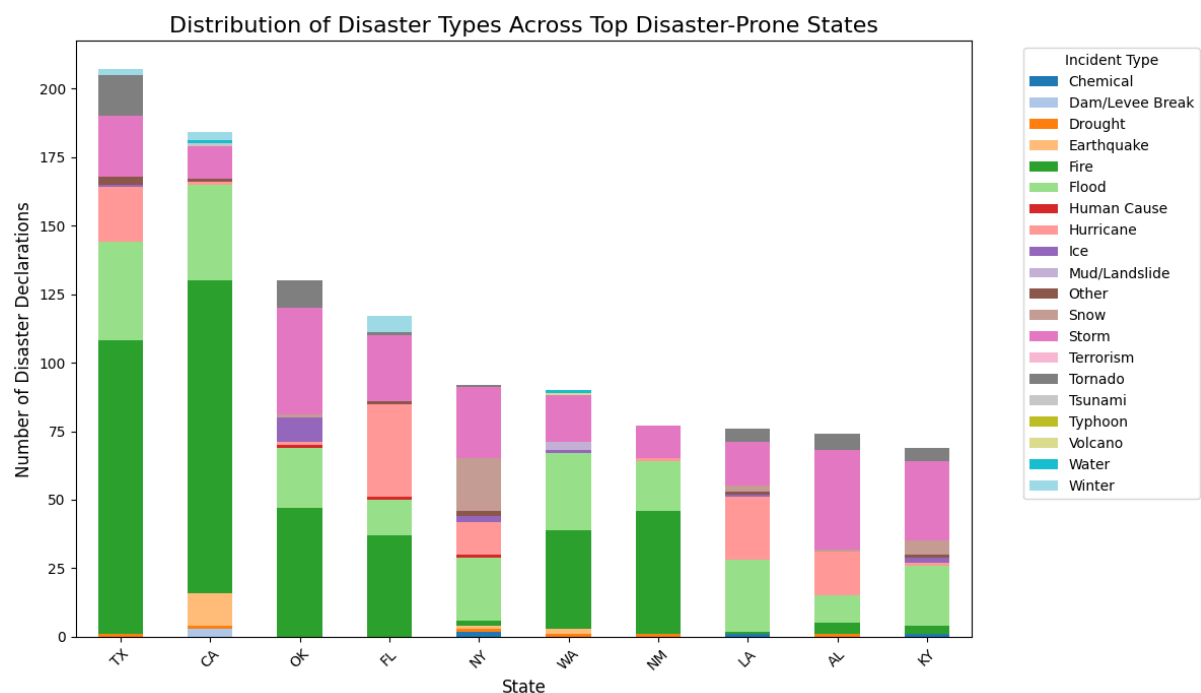
Incident Type Distribution by State :

Chart Type

Stacked Bar Chart

Description

Shows how different disaster types contribute to the total declarations within each state.



What the visualization represents

The stacked bar chart represents the distribution of different disaster types across the top disaster-prone states in the United States. Each bar corresponds to a state, and the different colored segments within the bar represent the various types of disasters such as fires, floods, hurricanes, storms, and tornadoes. The height of each segment shows how much each disaster type contributes to the total number of disaster declarations in that state.

What spatial pattern is observed

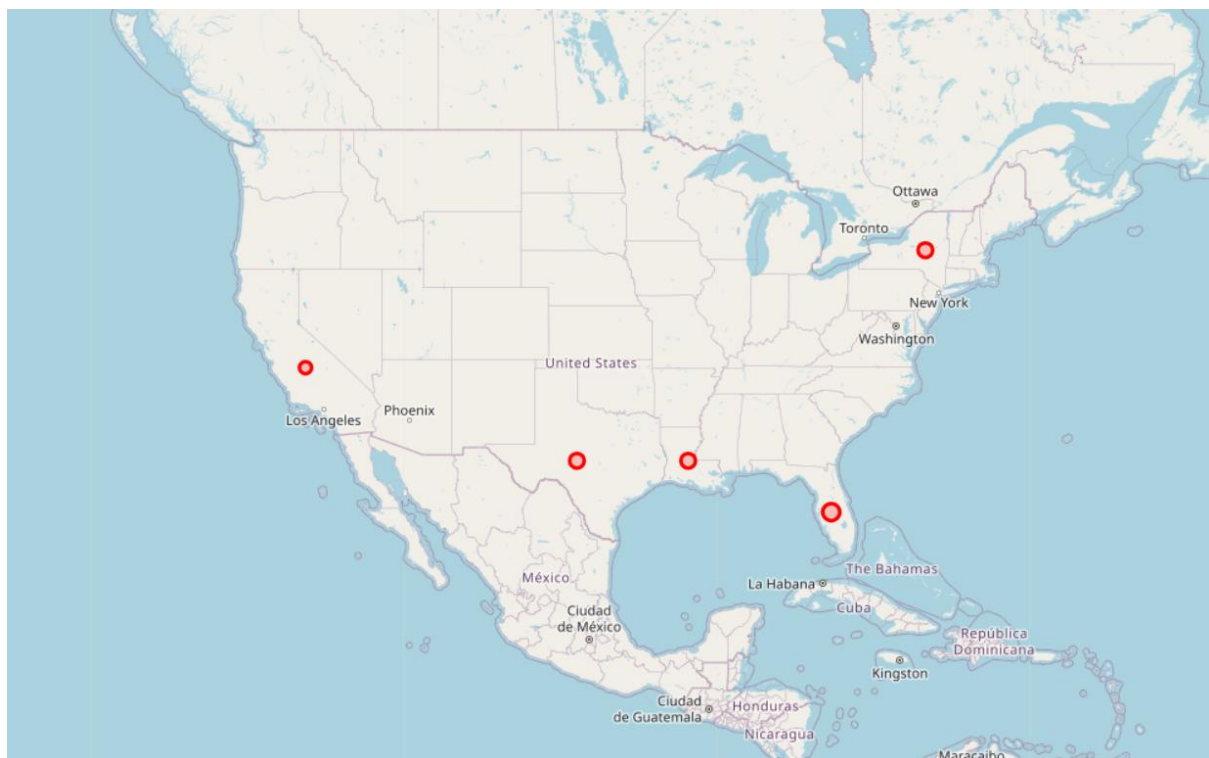
The visualization shows that different states are dominated by different disaster types. For example, fires contribute significantly to disaster declarations in western states like California and Texas, while hurricanes and floods appear more prominently in southern states such as Florida and Louisiana. Storm-related disasters also appear frequently across several states.

Possible explanation for the pattern

These patterns are largely influenced by regional climate and geographical conditions. Western states often experience wildfires due to dry climates, high temperatures, and dense vegetation, while southern and coastal states are more vulnerable to hurricanes, floods, and tropical storms due to their proximity to oceans and warm coastal waters. Additionally, some central states frequently experience severe storms and tornadoes, which contributes to the variety of disaster types observed in those regions.

Incident Type Hotspot Map :

Hurricane Hotspot:



What the visualization represents

The hurricane hotspot map shows how hurricane-related FEMA disaster declarations are distributed geographically across states. Each marker represents a state, and marker size/intensity reflects the number of hurricane declarations, so larger/darker markers indicate stronger hotspots.

What spatial pattern is observed

The strongest concentration is in coastal and Gulf-facing states. Florida appears as the most prominent hotspot (34 declarations), with Louisiana (23) and Texas (20) also showing high activity. This indicates that hurricane impacts are not evenly distributed and are clustered in vulnerable coastal regions.

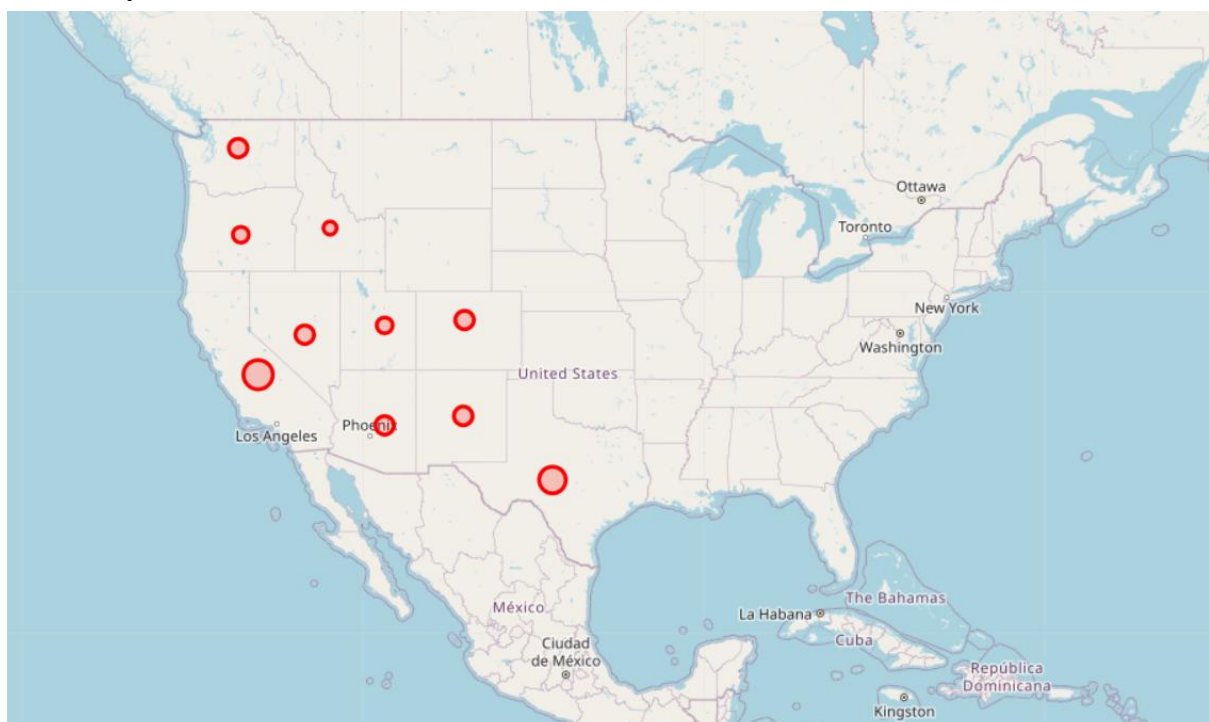
Possible explanation for the pattern

This pattern is driven by exposure to Atlantic and Gulf tropical cyclone tracks. States with long coastlines and warm nearby ocean waters face more frequent landfall and severe storm effects (wind, surge, flooding), which leads to repeated disaster declarations.

Key insight

Hurricane risk is regionally concentrated, especially in the Southeast and Gulf Coast, rather than spread uniformly across the U.S.

Fire Hotspots:



What the visualization represents

The fire hotspot map shows the geographic concentration of fire-related FEMA disaster declarations across U.S. states. Each marker represents a state, and marker size/intensity reflects the number of fire declarations, so larger/darker markers indicate stronger fire hotspots.

What spatial pattern is observed

The strongest hotspots are concentrated in western and southwestern states. California has the highest fire declarations (114), followed by Texas (107), with additional high counts in Oklahoma (47), New Mexico (45), Colorado (43), and Nevada (39). This indicates that fire risk is spatially clustered rather than evenly distributed.

Possible explanation for the pattern

This pattern is consistent with regional climate and land-cover conditions. Many western and southwestern states face hot, dry seasons, drought, and extensive wildland-urban interfaces, which increase wildfire likelihood and severity. Repeated dry-weather events and fuel buildup can lead to frequent fire-related declarations.

Conclusion:

In this milestone, geospatial and exploratory data analysis techniques were used to analyze the **distribution of natural disaster declarations across the United States**. By aggregating the dataset by state and incident type, several visualizations were created including a **choropleth map, state comparison bar chart, incident type distribution chart, and hotspot maps**.

The visualizations revealed clear geographical patterns in disaster occurrences. States such as **Texas, California, and Florida** show a significantly higher number of disaster declarations compared to other states. This indicates that certain regions of the United States are more vulnerable to natural disasters due to geographical and climatic conditions.

The analysis also showed that **different states experience different dominant disaster types**, such as hurricanes in coastal regions and wildfires in western states. These patterns help identify disaster-prone areas and improve understanding of regional disaster risks.

Overall, the geospatial visualizations successfully highlight **disaster hotspots, regional variations, and dominant disaster types**, demonstrating how spatial analysis can support better disaster preparedness, planning, and resource allocation.